QUALCOMM SNAPDRAGON AI LAB BUILD & PRESENT CHALLENGE

SENTINEL — Privacy-First Local Multimodal AI Companion

A privacy-first local multimodal AI companion that sees, reads, hears, remembers, understands, and helps users find information across their workspace.

SEE · READ · HEAR · REMEMBER · UNDERSTAND · FIND

100% On-Device Execution | Zero Cloud APIs | Qualcomm Snapdragon Copilot+ PC Target Architecture

Local Perception
Continuous desk object tracking via YOLO-X and EasyOCR text extraction.

Local Speech & Memory
Local Whisper STT (<800ms) with dense vector semantic search in SQLite.

Epistemic Fact-Checking
ResponseValidator separates confirmed observations from inferences.

Target Platform
Designed for Qualcomm Snapdragon X-series Windows Copilot+ PCs.

PROBLEM STATEMENT

The Fragmentation & Cloud Privacy Dilemma

Existing AI assistants fail in physical desktop environments due to architectural siloing and cloud dependency:

Modality Fragmentation
Vision tools don't remember context; chat models cannot see desk hardware; voice transcription lacks spatial coordinates.

Cloud Privacy Risk
Sending live desk cameras or confidential prototype schematics to cloud APIs can create enterprise security and compliance risks.

Network Latency & Connectivity
Cloud-dependent workflows introduce latency and require connectivity; offline operation is impossible.

Spatial Hallucination
Generic cloud LLMs guess object positions and invent document citations without physical grounding.

SOLUTION OVERVIEW

The Solution: Unified On-Device Physical Intelligence

SENTINEL integrates computer vision, document intelligence, voice understanding, and spatial memory into one unified local-first multimodal loop. Core components run on-device; Snapdragon QNN runtime path prepared for hardware validation.

SEE & TRACK
YOLO-X detection with deterministic IoU + centroid tracking, EMA smoothing. Verified on host CPU.

READ & REMEMBER
EasyOCR spatial binding + MiniLM 384-d vector embeddings with strict project isolation. Verified on host CPU.

UNDERSTAND
Evidence-aware multimodal orchestration with strict token budget management.

FIND & GROUND
Find It spatial localization returning structured HIGHLIGHT_OBJECT visual pulses.

SYSTEM ARCHITECTURE

Decoupled Architecture & Snapdragon Readiness

The application separates UI presentation from AI model runtimes via decoupled adapter boundaries:

Presentation Layer
Next.js 16 + React 19 + TailwindCSS cockpit with zero external analytics or network calls.

Orchestration Layer
SentinelOrchestrator coordinates input normalization, dynamic evidence collection, and validation.

AI Service Layer
VisionEngine, OCREngine, SpeechEngine, and MemoryRetriever operating locally.

Runtime Adapters
CPURuntimeAdapter (host development) <-> QNNRuntimeAdapter (Snapdragon Copilot+ PC target).

CORE DEMONSTRATION

Canonical Demo: 'Where is my water bottle?'

The primary competition demonstration showcases the complete physical-to-digital loop:

1. Natural Spoken Query
User asks by voice: 'Where is my water bottle?' Microphone captures only on hold.

2. Local STT (792 ms)
Whisper-Tiny transcribes speech entirely on host CPU (RTF 0.264x) without cloud offload.

3. Intent & Spatial Match
VoiceRouter delegates to FindItService, resolving target track_001 in 'center_right'.

4. Interactive Grounding
Returns HIGHLIGHT_OBJECT action; animated cyan pulse illuminates the bottle on camera.

REASONING & SAFETY

Epistemic Fact-Checking: Beyond Text Generation

SENTINEL enforces strict epistemic separation to eliminate hallucination in physical environments:

Observed Facts
Directly verified via active vision detections and OCR text regions on the desk.

Retrieved Facts
Verified source citations from SQLite memory notes or ingested PDF document chunks.

Inferred Hypotheses
Logical conclusions explicitly marked as deduction rather than direct camera observation.

Grounded Uncertainties
Explicitly reports unknown when evidence is absent rather than guessing.

PRIVACY & SECURITY

Privacy by Design: Zero Ambient Surveillance

Strict architectural privacy guarantees enforce complete user sovereignty over physical data:

Microphone Default OFF
Capture is strictly click-to-record; hardware handles release immediately upon release.

Camera Lifecycle Control
Camera tracking executes only while explicitly active; leaving page purges state.

Zero Media Persistence
Video frames and audio streams are processed in RAM; zero raw media saved to disk.

Anti-Surveillance
Categorically rejects facial recognition, biometric profiling, and ambient room monitoring.

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Snapdragon Deployment & Hardware Separation

Architectural separation preserves strict technical truthfulness between host and target profiles:

Development Host
12th Gen Intel Core i5-12450H (x86_64, Windows 11) running CPUExecutionProvider.

NPU Verification Status
Physical NPU execution is NOT VERIFIED on Intel host; zero fabricated metrics.

Snapdragon Target
Targeted for Snapdragon X-series Windows ARM64 PCs with 45 TOPS Hexagon NPU.

Optimization Stack
Qualcomm AI Engine Direct (QNN), QAIRT, and Qualcomm GenieX runtime architecture.

EMPIRICAL BENCHMARKS

Validation & Measured Host Telemetry

All metrics represent genuine, empirical measurements on the active workstation:

145 / 145 Tests Passed
Full backend test suite passing in 33.7s; 0 TypeScript errors; 0 lint warnings.

Whisper RTF = 0.2643
Real speech benchmark: 792.9 ms inference on 3.0s audio (>3.7x faster than real time).

YOLO-X: 88.34 ms
Real-time object detection with aspect-ratio letterboxing on local CPU.

MiniLM: 7.89 ms
384-dimensional query embedding with batch throughput of 273 items/sec.

COMPETITION CONCLUSION

Submission Readiness & What Makes SENTINEL Win

SENTINEL delivers a complete, grounded, reproducible submission candidate:

Clean Repository
Zero committed model binaries, zero secrets, and automated pre-flight verifier.

Complete Package
Official Brief Description (PDF), Pitch Deck (PPTX & PDF), and Canonical Demo Script.

Truthful Benchmarks
Host CPU results clearly documented; Snapdragon NPU targets cleanly separated.

Physical Innovation
True on-device physical AI that bridges desktop reality with digital memory.